GLOBAL COFFEE HEALTH ANALYSIS

Capstone #1

M2M Tech Connect
Data Talent Program
Cohort 18

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OBJECTIVE

- Analyze and describe the data from Global Coffee Health Dataset
- Visualize the data via charts using Python and Bokeh
- Visualize the data in a world map using Geo JSON and Bokeh
- Export the visualization to Netlify
- Tools used:
  - Google Colab Python Notebook
  - Pandas – data manipulation
  - Bokeh – data visualization

DATASET OVERVIEW

Source: https://www.kaggle.com/datasets/uom190346a/global-coffee-health-dataset

Num. of rows: 10,000

Num. of columns: 15

Notes:

- All columns are filled in except for "Health Issues" which have null values

```
Index: 10000 entries, 1 to 10000
Data columns (total 15 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Age                      10000 non-null  int64
 1   Gender                   10000 non-null  object
 2   Country                  10000 non-null  object
 3   Coffee_Intake            10000 non-null  float64
 4   Caffeine_mg              10000 non-null  float64
 5   Sleep_Hours              10000 non-null  float64
 6   Sleep_Quality            10000 non-null  object
 7   BMI                      10000 non-null  float64
 8   Heart_Rate               10000 non-null  int64
 9   Stress_Level             10000 non-null  object
 10  Physical_Activity_Hours  10000 non-null  float64
 11  Health_Issues            4059 non-null   object
 12  Occupation               10000 non-null  object
 13  Smoking                  10000 non-null  int64
 14  Alcohol_Consumption      10000 non-null  int64
dtypes: float64(5), int64(4), object(6)
memory usage: 1.2+ MB
```

DATASET OVERVIEW

Column	Type	Description
ID	Integer	Unique record ID (1–10000)
Age	Integer	Age of participant (18–80 years)
Gender	Categorical	Male, Female, Other
Country	Categorical	Country of residence (20 countries)
Coffee_Intake	Float	Daily coffee consumption in cups (0–10)
Caffeine_mg	Float	Estimated daily caffeine intake in mg **(1 cup ≈ 95 mg)**
Sleep_Hours	Float	Average hours of sleep per night (3–10 hours)
Sleep_Quality	Categorical	Poor, Fair, Good, Excellent (based on sleep hours)

Column	Type	Description
BMI	Float	Body Mass Index (15–40)
Heart_Rate	Integer	Resting heart rate (50–110 bpm)
Stress_Level	Categorical	Low, Medium, High (based on sleep hours and lifestyle)
Physical_Activity_Hours	Float	Weekly physical activity (0–15 hours)
Health_Issues	Categorical	None, Mild, Moderate, Severe (based on age, BMI, and sleep)
Occupation	Categorical	Office, Healthcare, Student, Service, Other
Smoking	Boolean	0 = No, 1 = Yes
Alcohol_Consumption	Boolean	0 = No, 1 = Yes

CLEANING THE DATA

- Drop the columns ‘Smoking’ and ‘Alcohol Consumption’ using data frame DROP function
- Preview (First 5 rows and Last 5 Rows)

```
df_coffee.head()
```

1 to 5 of 5 entries Filter

ID	Age	Gender	Country	Coffee_Intake	Caffeine_mg	Sleep_Hours	Sleep_Quality	BMI	Heart_Rate	Stress_Level	Physical_Activity_Hours	Health_Issues	Occupation
1	40	Male	Germany	3.5	328.1	7.5	Good	24.9	78	Low	14.5	NaN	Other
2	33	Male	Germany	1.0	94.1	6.2	Good	20.0	67	Low	11.0	NaN	Service
3	42	Male	Brazil	5.3	503.7	5.9	Fair	22.7	59	Medium	11.2	Mild	Office
4	53	Male	Germany	2.6	249.2	7.3	Good	24.7	71	Low	6.6	Mild	Other
5	32	Female	Spain	3.1	298.0	5.3	Fair	24.1	76	Medium	8.5	Mild	Student

```
df_coffee.tail()
```

1 to 5 of 5 entries Filter

ID	Age	Gender	Country	Coffee_Intake	Caffeine_mg	Sleep_Hours	Sleep_Quality	BMI	Heart_Rate	Stress_Level	Physical_Activity_Hours	Health_Issues	Occupation
9996	50	Female	Japan	2.1	199.8	6.0	Fair	30.5	50	Medium	10.1	Moderate	Healthcare
9997	18	Female	UK	3.4	319.2	5.8	Fair	19.1	71	Medium	11.6	Mild	Service
9998	26	Male	China	1.6	153.4	7.1	Good	25.1	66	Low	13.7	NaN	Student
9999	40	Female	Finland	3.4	327.1	7.0	Good	19.3	80	Low	0.1	NaN	Student
10000	42	Female	Brazil	2.9	277.5	6.4	Good	28.1	72	Low	9.8	NaN	Student

FILL NULL VALUES

- ‘Health_Issues’ column has 4,059 non-null values, which means it has 5,941 null values
- Fill null values with ‘No Issues’ using FILLNA function on the data frame column

```
df_coffee['Health_Issues'] = df_coffee['Health_Issues'].fillna('No Issues')
df_coffee.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 10000 entries, 1 to 10000
Data columns (total 13 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Age                      10000 non-null  int64  
 1   Gender                   10000 non-null  object 
 2   Country                  10000 non-null  object 
 3   Coffee_Intake            10000 non-null  float64
 4   Caffeine_mg              10000 non-null  float64
 5   Sleep_Hours              10000 non-null  float64
 6   Sleep_Quality            10000 non-null  object 
 7   BMI                      10000 non-null  float64
 8   Heart_Rate               10000 non-null  int64  
 9   Stress_Level             10000 non-null  object 
 10  Physical_Activity_Hours  10000 non-null  float64
 11  Health_Issues            10000 non-null  object 
 12  Occupation               10000 non-null  object 
dtypes: float64(5), int64(2), object(6)
memory usage: 1.1+ MB
```

DESCRIBE THE DATA FRAME

- The DESCRIBE function of the data frame will show us statistical values for each column like average, max value, min value and the percentiles.

```
1 df_coffee.describe()
```

1 to 8 of 8 entries Filter

index	Age	Coffee_Intake	Caffeine_mg	Sleep_Hours	BMI	Heart_Rate	Physical_Activity_Hours
count	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0
mean	34.9491	2.50923	238.41101	6.63622	23.986859999999997	70.6178	7.487040000000001
std	11.160939290516124	1.450247609518068	137.74881462057328	1.2220554216341164	3.9064113128720503	9.82295129771216	4.315179963363283
min	18.0	0.0	0.0	3.0	15.0	50.0	0.0
25%	26.0	1.5	138.75	5.8	21.3	64.0	3.7
50%	34.0	2.5	235.4	6.6	24.0	71.0	7.5
75%	43.0	3.5	332.025	7.5	26.6	77.0	11.2
max	80.0	8.2	780.3	10.0	38.2	109.0	15.0

DATA VISUALIZATION
AGE VS CAFFEINE INTAKE (MG)

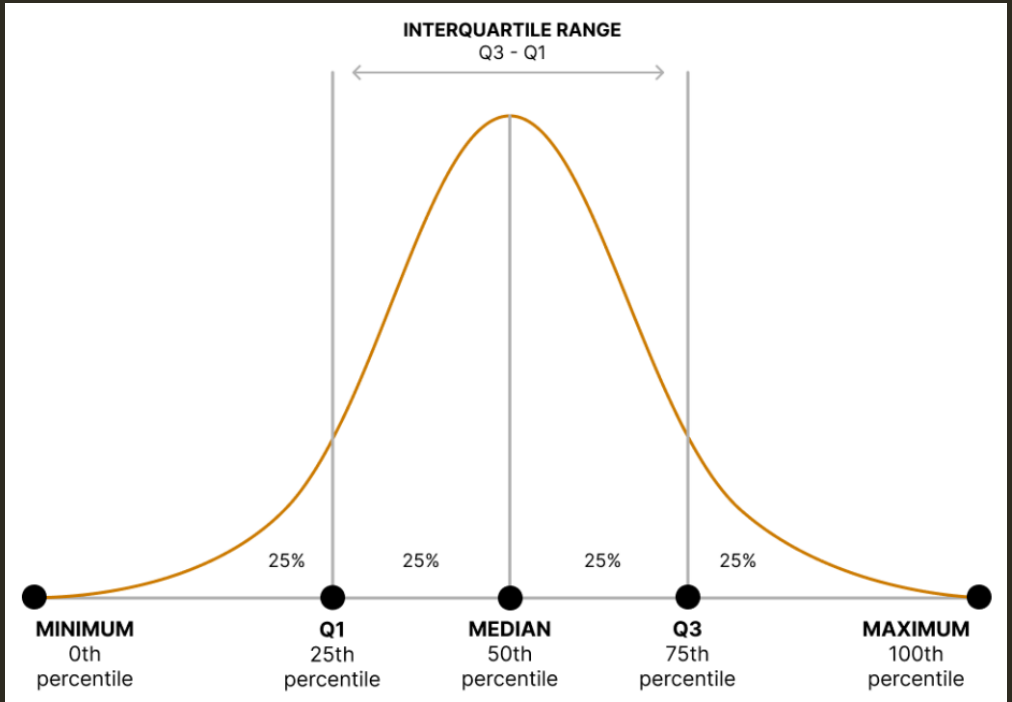


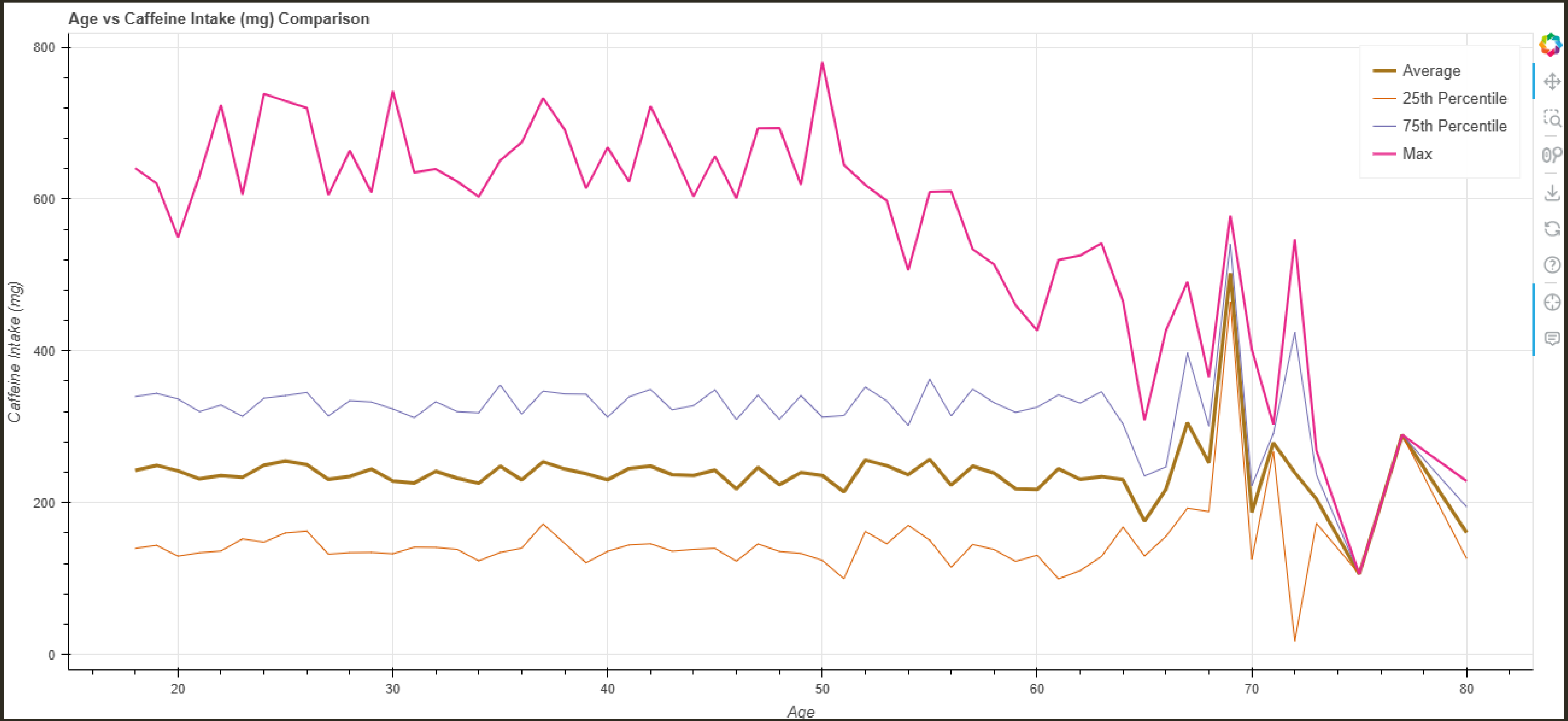
Image Source:
https://comet.arts.ubc.ca/docs/2_Beginner/beginner_central_tendency/beginner_central_tendency.html

- Get the Caffeine Intake (mg) consumed by each individual based on their Age
- For Caffeine Intake (mg), we will be showing the following for each Age:
  - Average – is a single number that best represents a set of data. It will represent how much Caffeine Intake each Age would most likely have
  - Max – the highest amount of Caffeine Intake (mg) by an individual in that Age
  - 25th Percentile – is the point in a dataset below which 25% of the data values fall, meaning that 75% of the data values are higher than this point. Example: If those who are age 20 take 129 mg of Caffeine in the 25th Percentile, that means 25% of them takes less than 129 mg while 75% of them takes more than 129 mg.
  - 75th Percentile – is the point in a dataset below which 75% of the data values fall, meaning that 25% of the data values are higher than this point. Example: If those who are age 20 take 336 mg of Caffeine in the 75th Percentile, that means 75% of them takes less than 336 mg while 25% of them takes more than 336 mg.

AGE VS CAFFEINE INTAKE (MG) CODE PROCESS

1. Group by Age column
2. Create functions to calculate 25^{th} and 75^{th} percentile named q25 and q75
3. Call AGGREGATE function on group's Caffeine_mg column. AGGREGATE will get the mean, max, size and call the 25^{th} and 75^{th} percentile functions
4. AGGREGATE will create a new data frame with the following columns (Age – index, mean, size, max, q25, q75)
5. Define a Column Data Source for each AGGREGATE data (mean, size, max, 25%, 75%)
6. Create Bokeh figure and add a Line for each AGGREGATE data (mean, size, max, 25%, 75%)
7. Add Hover Tool and Crosshair Tool
8. Show the figure

DATA VISUALIZATION
AGE VS CAFFEINE INTAKE (MG)



DATA VISUALIZATION
AGE GROUP AND GENDER VS CAFFEINE INTAKE (MG)

- Get the Caffeine Intake (mg) consumed by each individual based on each Gender on each Age Group
- The Age Group will be determined by decade as shown in the table on the right
- The data will be grouped by Age Group, and each Age Group will contain each unique Gender (Male, Female, Other)

Age Group	Ages Covered
10s (teens)	10 – 19 yrs. old
20s	20 – 29 yrs. old
30s	30 – 39 yrs. old
40s	40 – 49 yrs. old
50s	50 – 59 yrs. old
60s	60 – 69 yrs. old
70s	70 – 79 yrs. old
80s	80 – 89 yrs. old

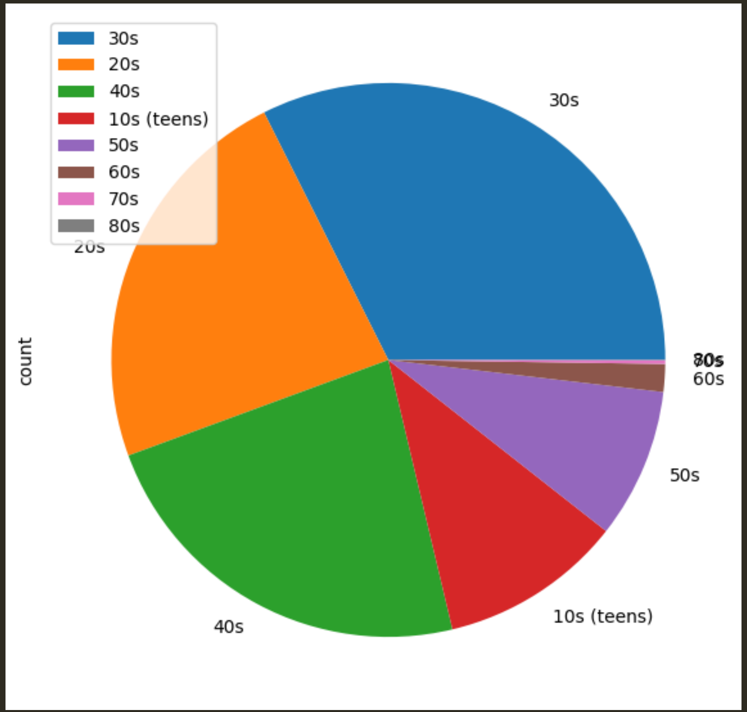
FORMING THE AGE GROUP

Create a function that will return the Age Group based on the Age

Apply the function to the Age column in the data frame

Assign the result to a new data frame column named Age Group

Age Group	Ages Covered
10s (teens)	10 – 19 yrs. old
20s	20 – 29 yrs. old
30s	30 – 39 yrs. old
40s	40 – 49 yrs. old
50s	50 – 59 yrs. old
60s	60 – 69 yrs. old
70s	70 – 79 yrs. old
80s	80 – 89 yrs. old

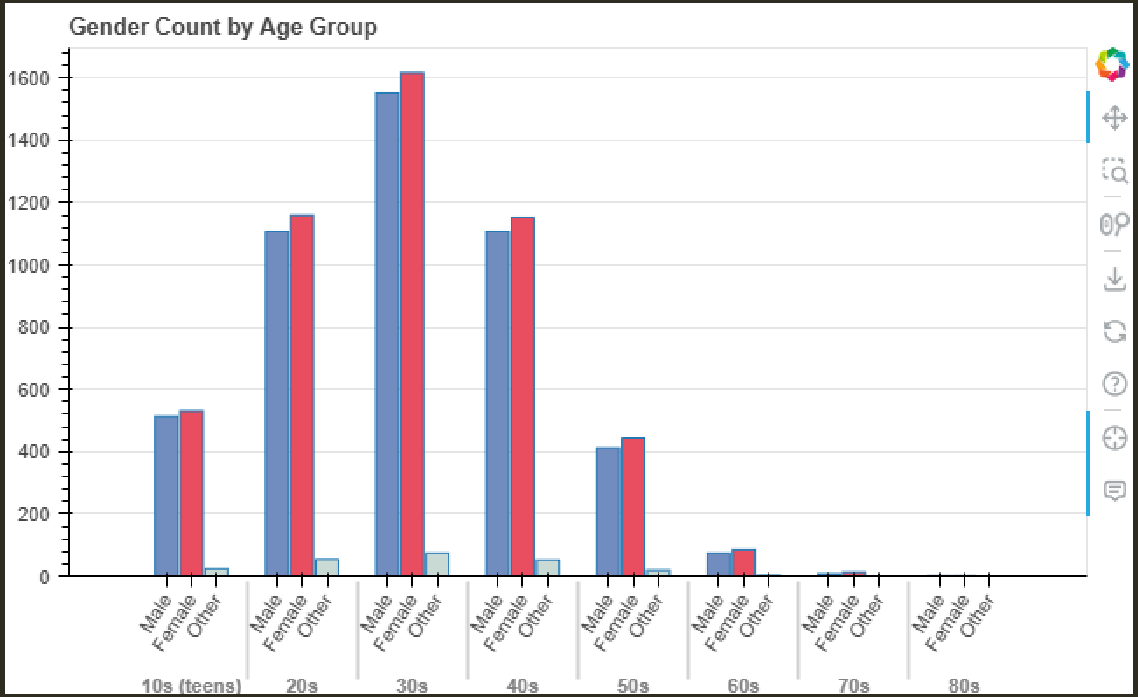


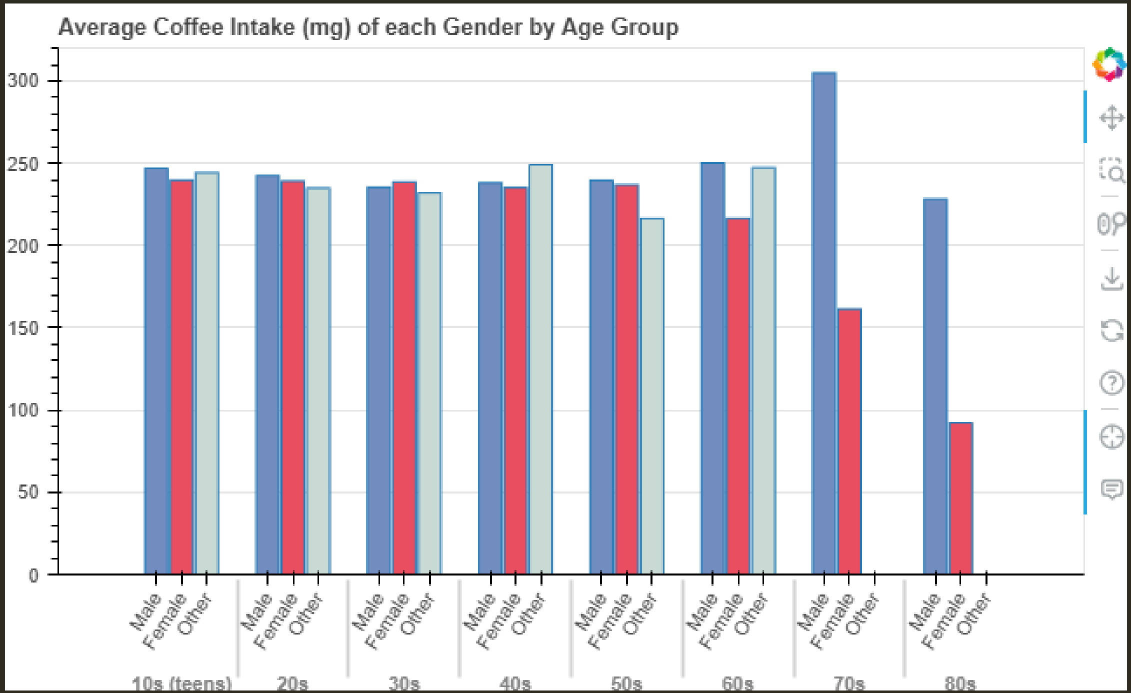

```
age_group_count = df_coffee['Age_Group'].value_counts()
age_group_count.plot.pie(legend=True, figsize=(7,7))
print(age_group_count)
```

```
Age_Group
30s            3241
20s            2320
40s            2311
10s (teens)    1068
50s             875
60s             162
70s              21
80s               2
Name: count, dtype: int64
```

AGE GROUP AND GENDER VS CAFFEINE INTAKE (MG) CODE PROCESS

1. Form the Age Group column in the data frame
2. Get the Unique Genders and store them in an array (Male, Female, Other)
3. For each Gender, create a group and group them by Age Group
4. For each Gender group, call AGGREGATE on the Caffeine_mg column and get the mean and size
5. Calling AGGREGATE on each group will create a new data frame with the following columns (Age Group – index, mean, size)
6. **Clean the Data Frames.** If the data frame has missing Age Group due to having no data, manually add rows for those Age Group
7. **Prepare the data.** Create an array that contains the age and gender pairs so that we will have every gender for each age group
8. Create an array that contains all the **size** data for each age-gender pair
9. Create an array that contains all the **mean** data for each age-gender pair
10. Define a Column Data Source that contains our age-gender pair array, the size data array and the mean data array
11. Create two Bokeh figures, one for the size data, one for the mean data
12. Add a Vertical Bar on each figure referencing the Column Data Source
13. Add Hover Tool and Crosshair Tool
14. Show the figure



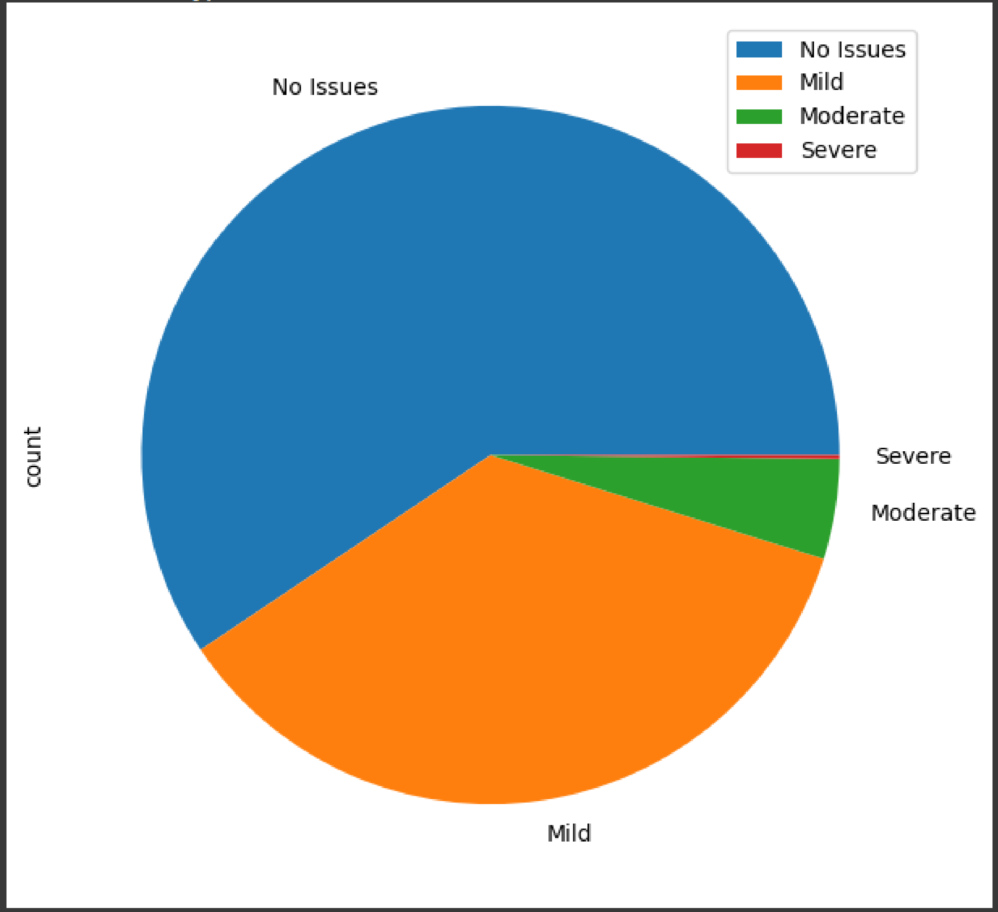


DATA VISUALIZATION
AGE GROUP AND GENDER VS CAFFEINE INTAKE (MG)

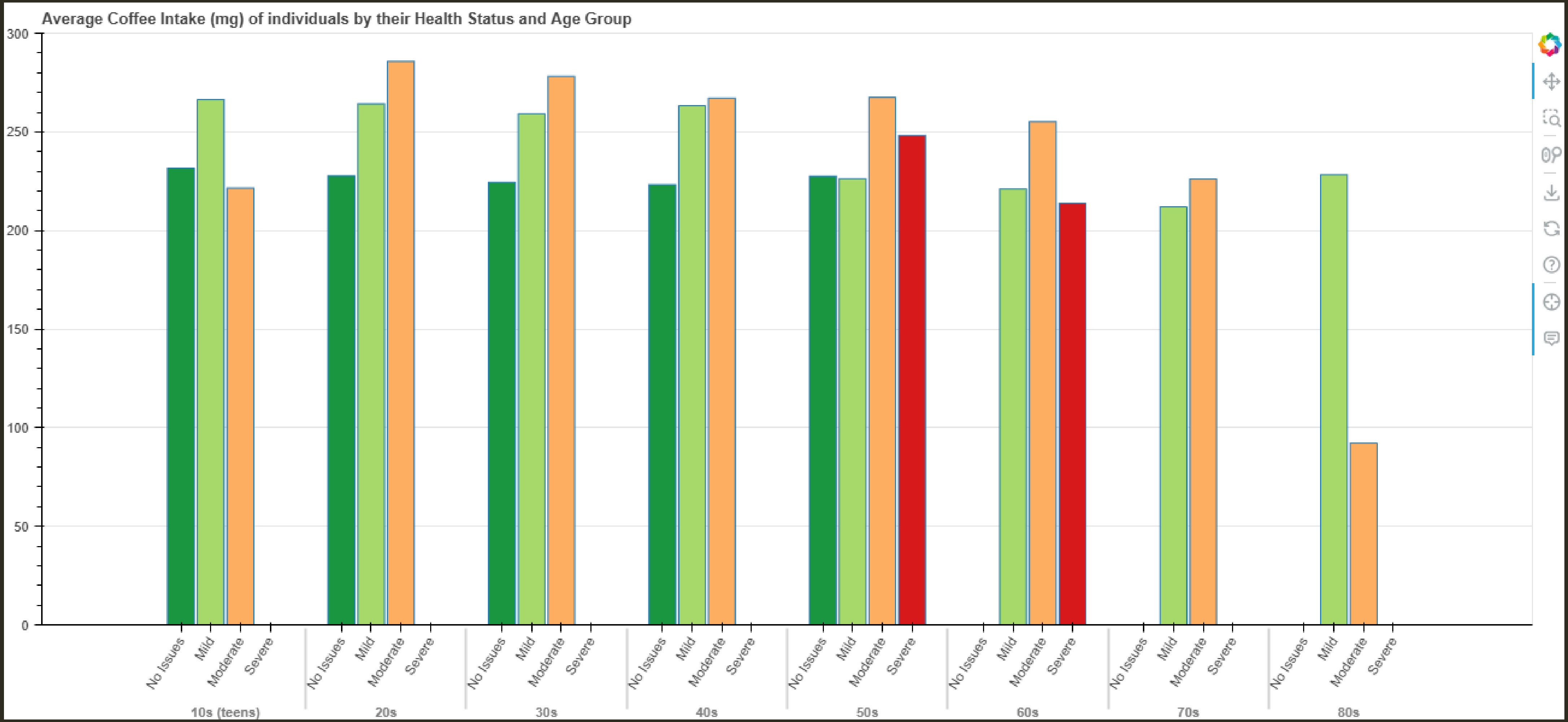
DATA VISUALIZATION AGE GROUP AND HEALTH ISSUES VS CAFFEINE INTAKE (MG)

- Get the Caffeine Intake (mg) consumed by each individual based on each Health Issue category on each Age Group
- Similar to the previous visualization, the Age Group will be determined by decade
- The data will be grouped by Age Group, and each Age Group will contain each unique Health Issue category (No Issues, Mild Moderate, Severe)
- The plot pie on the right shows the count of each Health Issue category in our data set

```
Health_Issues
No Issues    5941
Mild         3579
Moderate      463
Severe         17
Name: count, dtype: int64
```

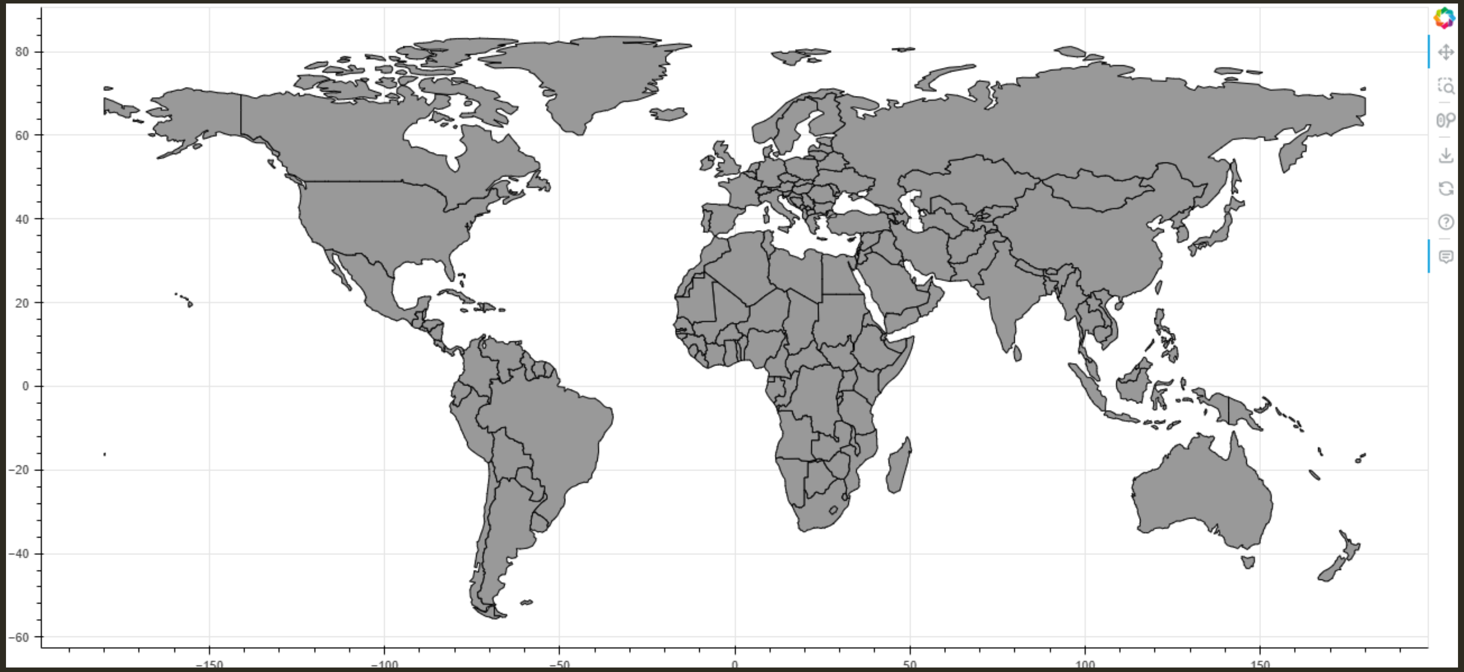


DATA VISUALIZATION
AGE GROUP AND HEALTH ISSUES VS CAFFEINE INTAKE (MG)



DATA VISUALIZATION AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY

- Get the average Caffeine Intake (mg) consumed by each Country and visualize using a world map
- The world map will be shown using Geo JSON
- We will get the world map Geo JSON file from the following source: https://geojson-maps.kyd.au/
- Shown on the right is how the world map will look like after importing the Geo JSON file and showing it to a Bokeh plot
- Our goal is to change the color of the countries that has existing data and add a hover tool for more details



INVOLVED COUNTRIES

- When we group our data frame by Country, we will see how many countries we have data on
- After calling SIZE on the group object, we will see that we have 20 countries
- We should expect a lot of grey areas in our visualization since we are only highlighting 20 countries
- Note that UK and USA spellings are abbreviated

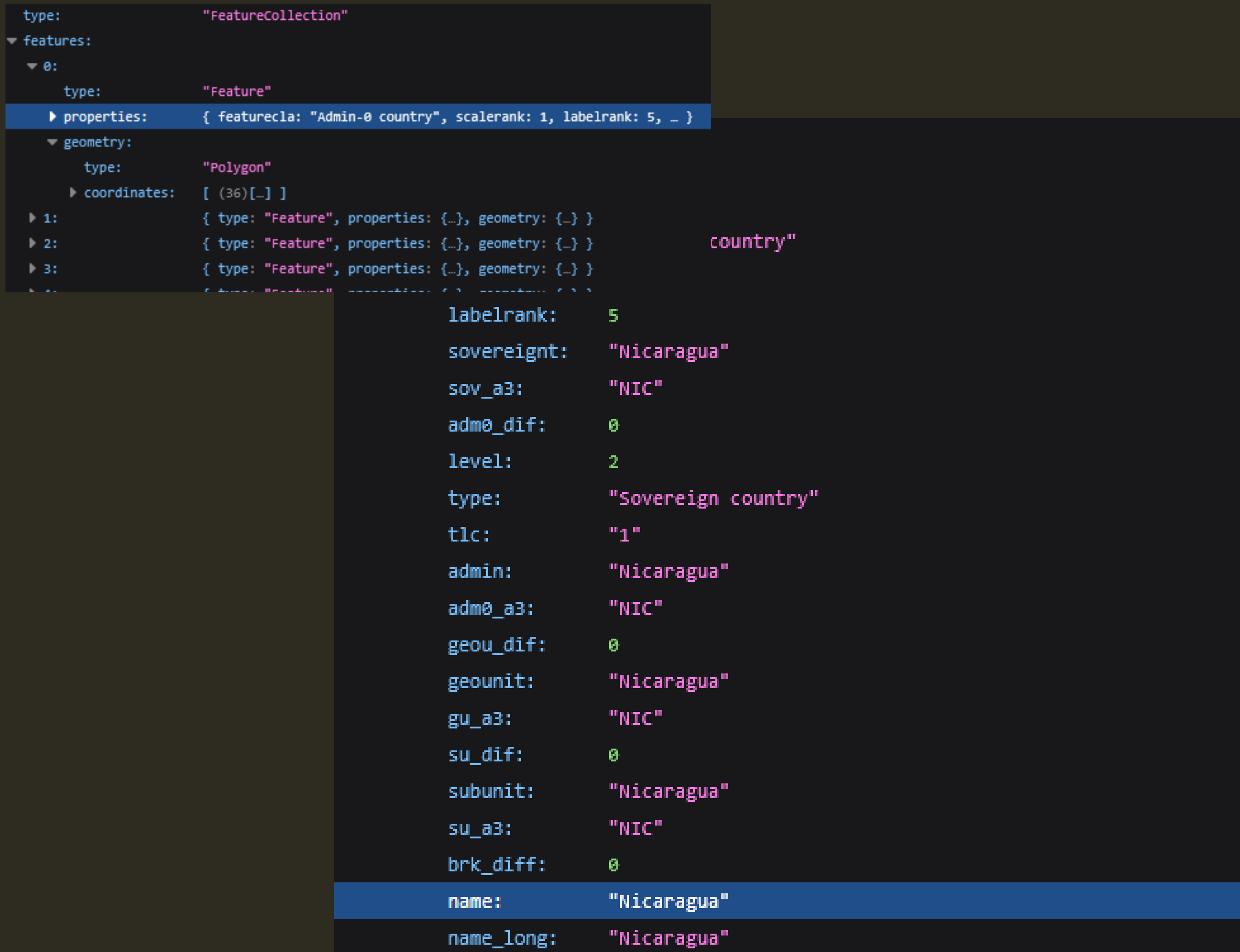
```
world_country_group = df_coffee.groupby('Country')
print('Number of groups', world_country_group.ngroups,'\n')
print(world_country_group.size())
```

```
Number of groups 20

Country
Australia      497
Belgium        497
Brazil         456
Canada         543
China          521
Finland        510
France         499
Germany        497
India          524
Italy          509
Japan          469
Mexico         483
Netherlands    494
Norway         523
South Korea    512
Spain          486
Sweden         513
Switzerland    500
UK             519
USA            448
dtype: int64
```

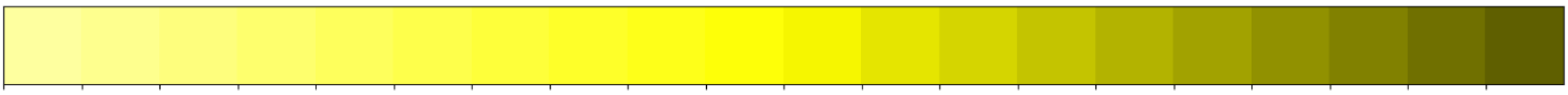
EXPLORING THE WORLD MAP’S JSON FILE

- The world map’s JSON file that we imported is a type of ‘Feature Collection’ and contains the following layers:
  - [‘features’] will be a list of features, with a total of 175 items representing each country
  - Each feature will contain ‘type’, ‘properties’ and ‘geometry’ objects
    - [‘properties’] is a very long dictionary containing the properties of the feature. We will be needing the ‘name’ property to match our data. We will also modify the dictionary to add more properties such as Color, Rank and the Average Caffeine Intake
    - [‘geometry’] is a Polygon type which Bokeh uses to draw our world map plot, we won’t be touching this
- Since we know how the JSON file looks like, we will access the properties in Python like this:
  - Geojson[‘features’][i][‘properties’]
    - ‘i’ will be the index in the ‘features’ list



DETERMINE THE COLOR BASED ON RANKING

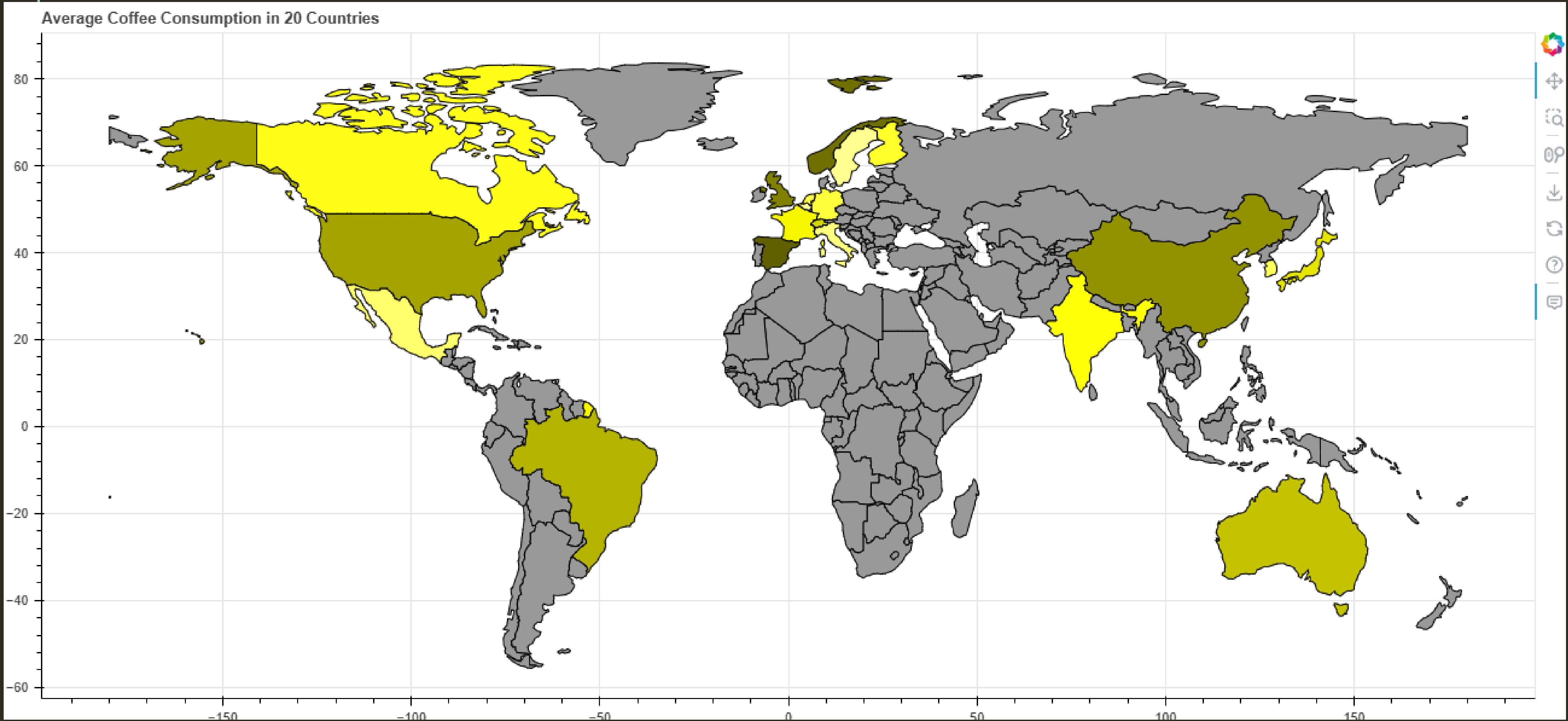
- To color our world map, we will be using Yellow as a theme
- Our goal is to change the color's darkness based on the Countries ranking based on how much average Caffeine intake they have
- The higher the Caffeine intake, the darker the Yellow will be
- The lower the Caffeine intake, the lighter the Yellow will be
- We will accomplish this by making a new function where it will take an amount parameter that will determine the color's darkness
- After testing the function, the colors will look like the ones shown below using 'seaborn' and its PALPLOT method



AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY CODE PROCESS

1. Import the world map JSON file to GeoJSON data source object
2. Group the data frame by Country
3. Prepare the Data. Rename USA and UK to their full name
4. Using the group object, get the mean of the Caffeine_mg column in ascending order
5. Getting the mean will create a new data frame containing the Country as the index and the mean of Caffeine_mg of each Country
6. Create a function that will give us a color where its lightness or darkness will depend on a supplied parameter
7. Add a column in the mean data frame that will contain the color of the Country using the new function
8. Add a column in the mean data frame that will determine its ranking/order based on the mean
9. Update Geo JSON properties to add new properties for color, rank and the mean
10. Create the Bokeh figure and add tooltips displaying the Country name, rank, and the mean
11. Show the map figure

DATA VISUALIZATION
AVERAGE CAFFEINE INTAKE (MG) BY COUNTRY



PREPARE TO PUBLISH

- In total we have shown 5 Bokeh figures
- To show all these figures into one, we will use Bokeh's layout functionalities: https://docs.bokeh.org/en/latest/docs/user_guide/basic/layouts.html#general-grid-layout
- Then we will use Bokeh's output file to export the figures in an HTML file
- The platform that we will be using to publish our web page is Netlify:
  - https://aramirez-capstone1-0978da.netlify.app/

```python
from bokeh.layouts import layout
from bokeh.plotting import output_file

final_layout = layout([
    [p_age_gender_size, p_age_gender_avgCaf],
    [p_age_avgCaf],
    [p_age_health_avgCaf],
    [p_map]
])

output_file('index.html')

show(final_layout)
```

ANALYSIS

- Age vs Caffeine Intake (mg) Comparison
  - The maximum Caffeine Intake in mg is 780.3 consumed by one of the individuals of age 50
  - From ages 18 to 64, the average Caffeine Intake is around 250 mg or 2.5 cups a day, the lower 25% consumes around less than 130 mg a day, and the upper 25% consumes more than 330 mg a day
  - After age 65, our data sample size lessens, which causes varying averages, max values and percentiles
- Age Group and Gender vs Caffeine Intake (mg) Comparison
  - The Age Group with most data is the 30s group (30-39 yrs)
  - Male to Female comparisons are almost the same where Female are slightly more. We do not have much data for the Other gender group
  - Average Caffeine consumption is no more than 250 mg a day for all genders, starting from teen until the 60s Age Group
- Age Group and Health Issue Category vs Caffeine Intake (mg) Comparison
  - The 'Severe' Health Issue Category does not have enough sample size, 11 in the 50s age group and 6 in the 60s age group
  - From Age Groups 20s to 40s, those who consume more than 250 mg have Mild to Moderate Health Issues
  - Those who does not have Health Issues consumes around 200 mg to 250 mg
  - The difference in consumption of those with no health issues to those that have is only around 50 mg, which suggests that Coffee consumption might have less impact on one's health, however, consuming less than 250 mg would be a safe measure
- Average Coffee Consumption by Country
  - Spain ranks number 1 with an average of 247 mg per day
  - Belgium ranks last at number 20 with an average of 230 mg per day

SOURCES

Dataset
https://www.kaggle.com/datasets/uom190346a/global-coffee-health-dataset
Geo JSON of World Map
https://geojson-maps.kyd.au/
Coding Help
Get percentile - https://stackoverflow.com/a/57087989
Lighten Color - https://stackoverflow.com/questions/37765197/darken-or-lighten-a-color-in-matplotlib
Bokeh Grid Layout - https://docs.bokeh.org/en/latest/docs/user_guide/basic/layouts.html#general-grid-layout


THANK YOU